CEMADEN

Centro Nacional de Monitoramento e Alertas de Desastres Naturais

# Monitoramento e Previsões da Situação do rio Tocantins

Outubro de 2026

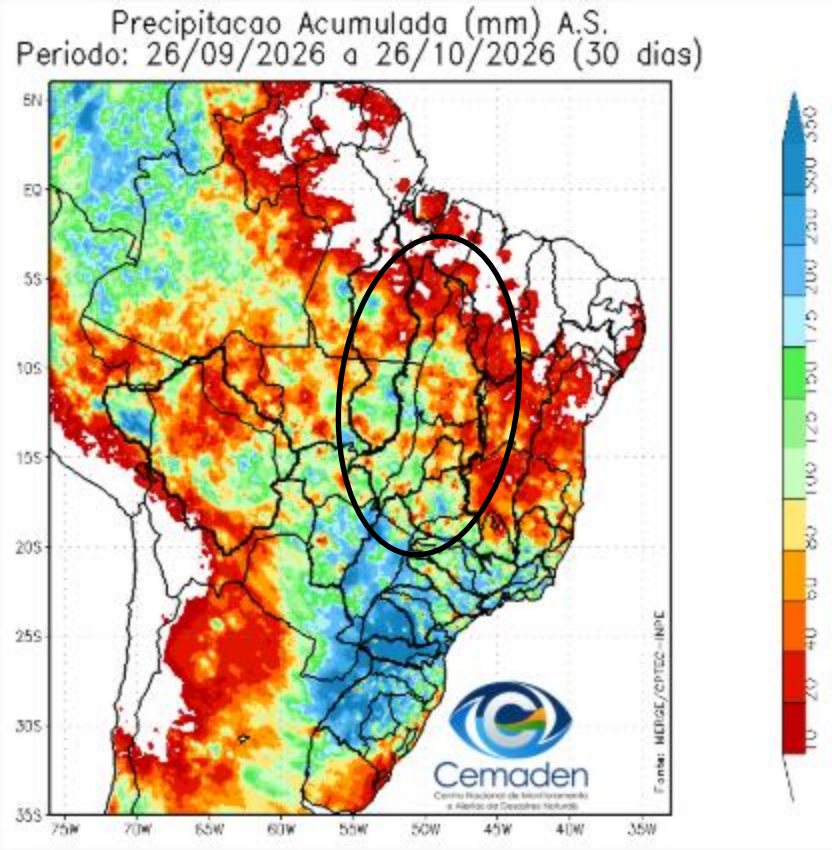


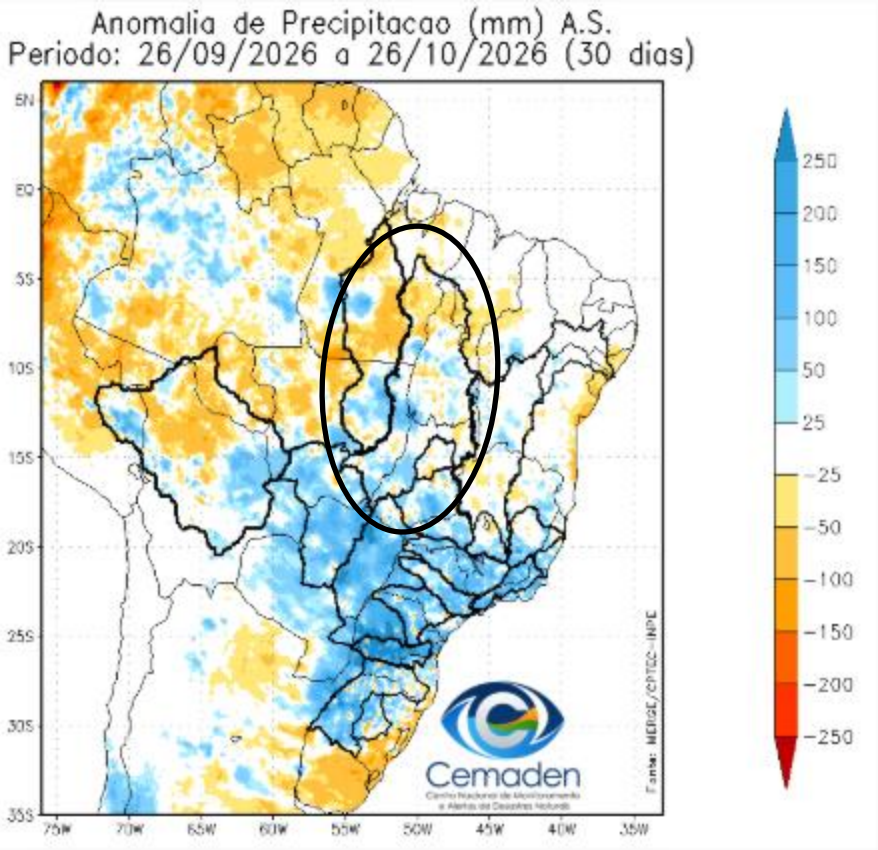
Cemaden

MINISTÉRIO DA CIÊNCIA, TECNOLOGIA E INOVAÇÃO

GOVERNO FEDERAL BRASIL UNIÃO E RECONSTRUÇÃO

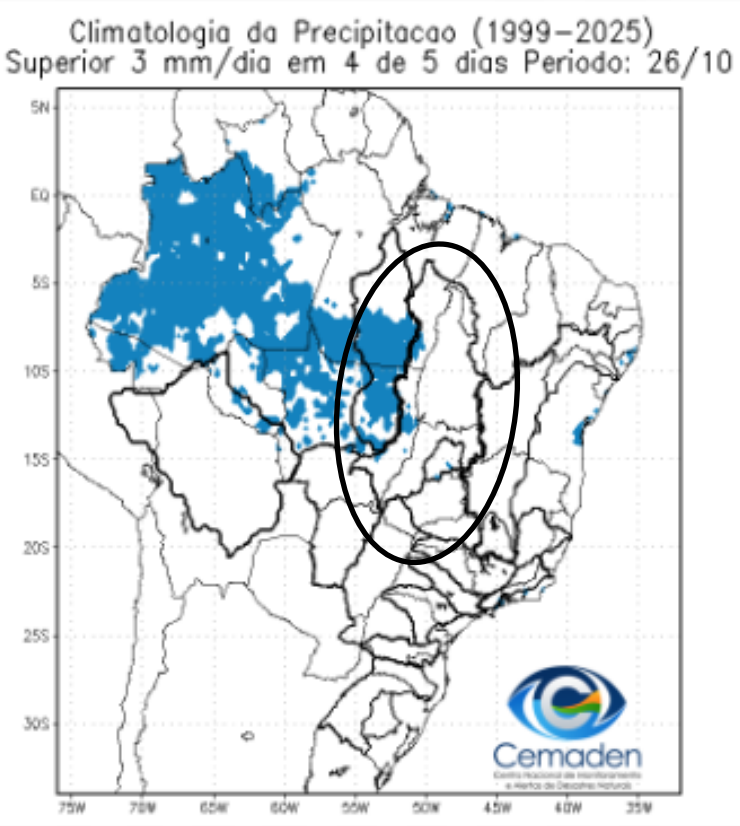
# Precipitação acumulada últimos 30 dias

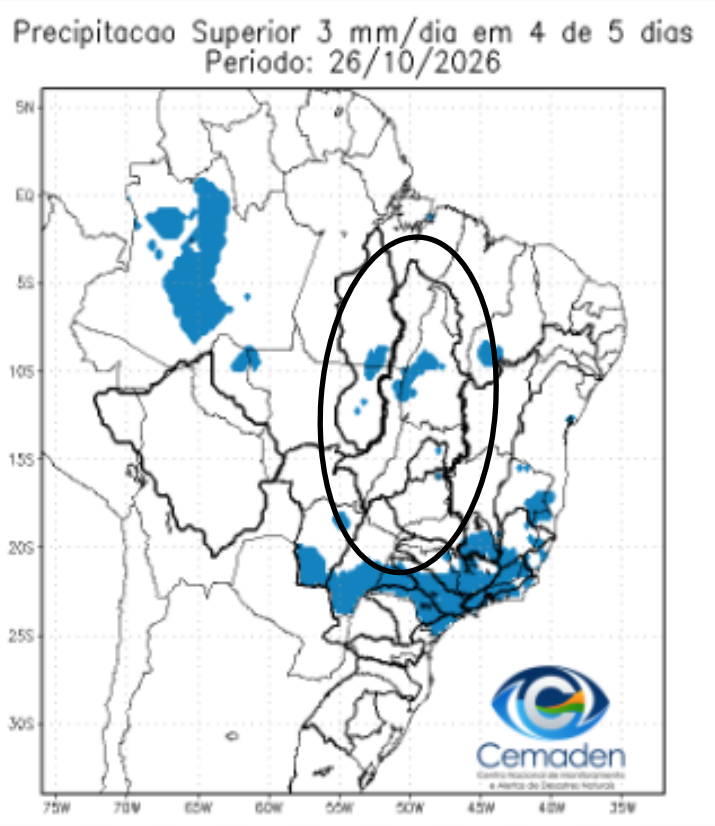



Fonte dos dados: Merge/INPE

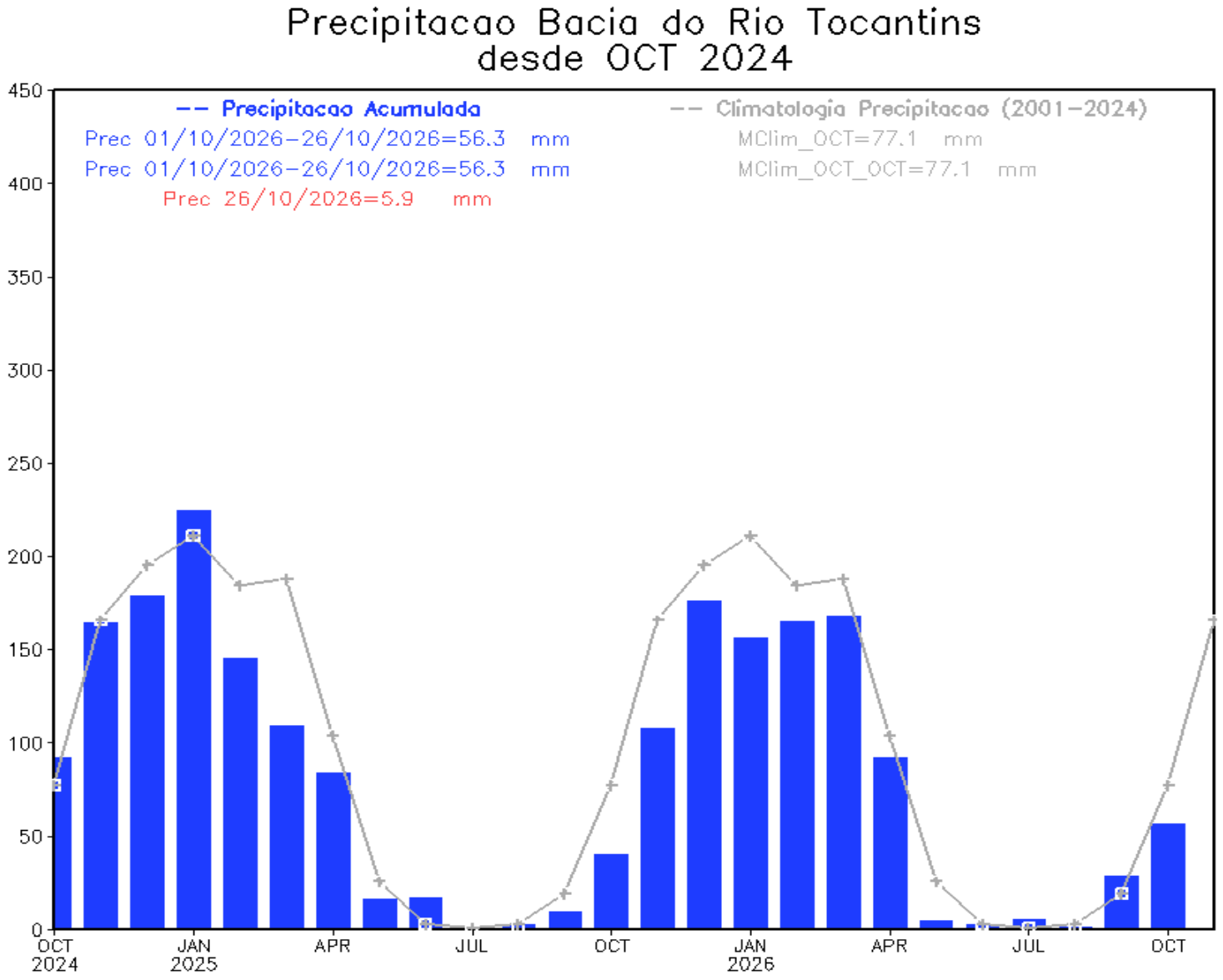
# Situação da Estação Chuvosa



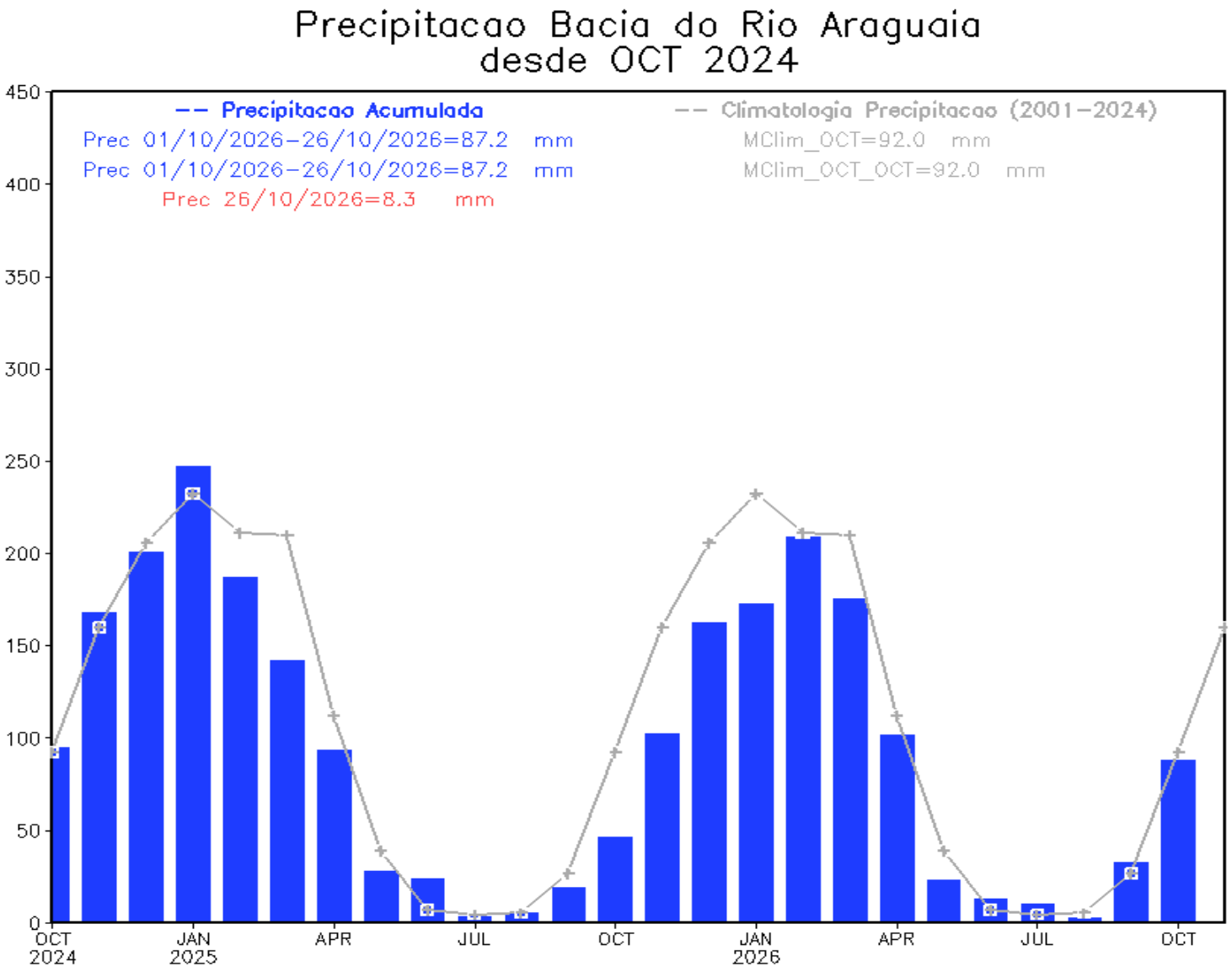

Fonte dos dados: Merge/INPE

# Precipitação Acumulada nos últimos Dois Anos



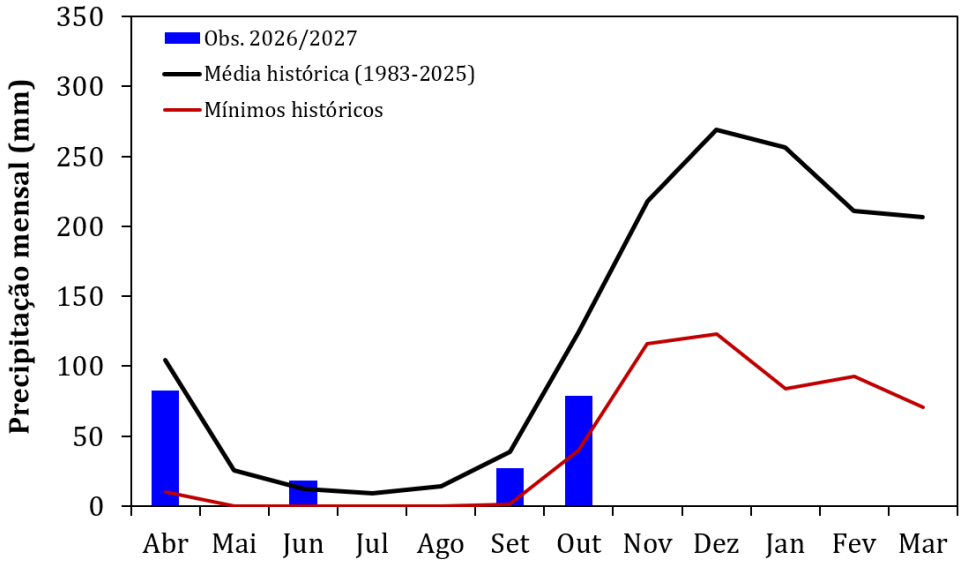

Fonte dos dados: Merge/INPE

# Precipitação Acumulada nos últimos Dois Anos




Fonte dos dados: Merge/INPE

# Monitoramento UHE Serra da Mesa



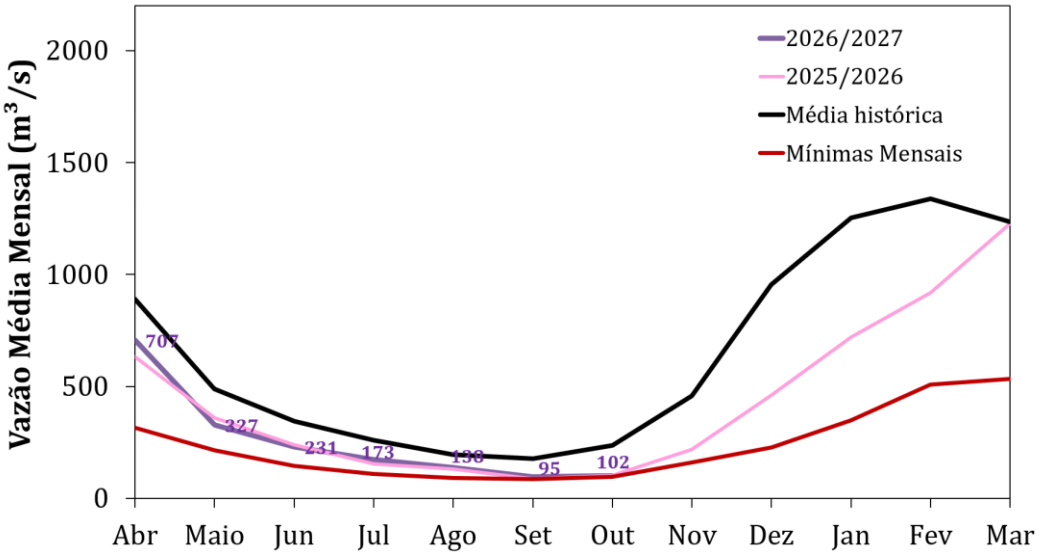

## Precipitação

**Estação Chuvosa - Out a Abr – 1390 mm**
2024/2025: 1355 mm **(97% da MLT)**
2025/2026: 1260 mm **(91% da MLT)**

**Estação Seca - Maio a Set – 100 mm**
2025: 24 mm **(23% da MLT)**
2026: 46 mm **(46% da MLT)**

**Out/2026*: 79** mm **(64% da MLT)**

**Até 25/10/2026*




## Vazão

**Estação Chuvosa - Out a Abr – 909 m³/s**
2024/2025: 881 m³/s **(97% da MLT)**
2025/2026**: 622 m³/s **(68% da MLT)**

**Estação Seca - Maio a Set – 293 m³/s**
2025: 194 m³/s **(66% da MLT)**
2026: 193 m³/s **(66% da MLT)**

**Out/26**: 102** m³/s **(43% da MLT)**
**25/Out: 72** m³/s **(30% da MLT)**

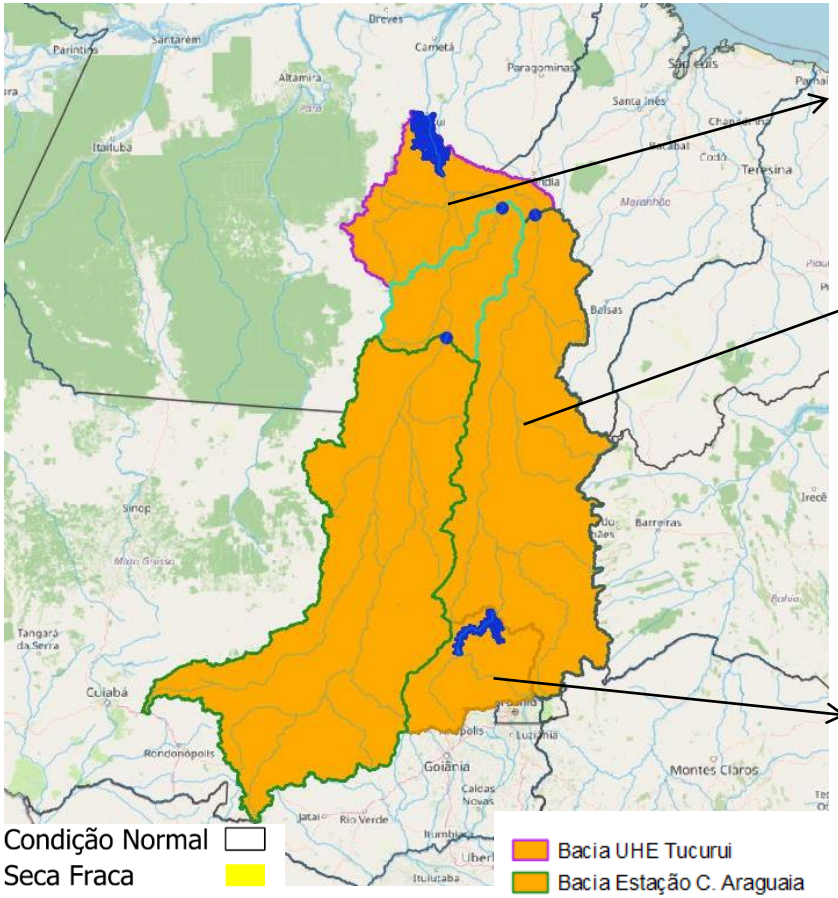
***Até 25/10/2026*

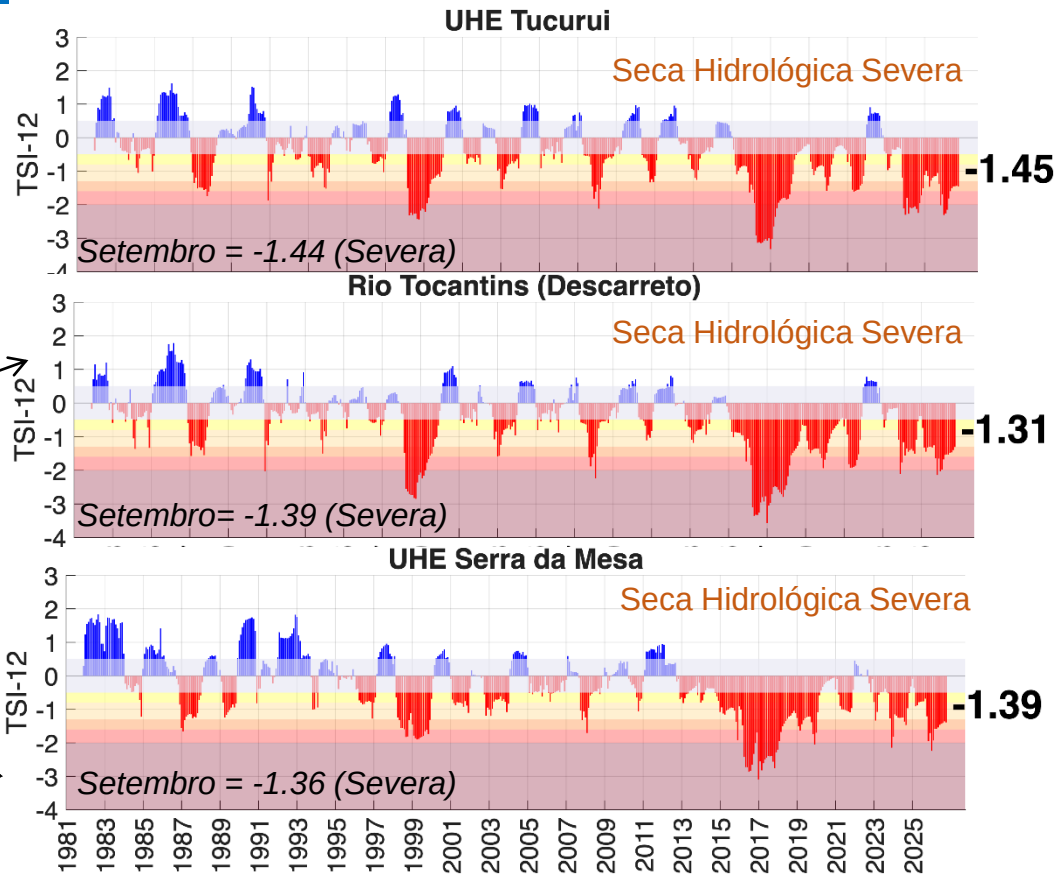
***Média de Longo Termo – MLT = 1983-2025***

Dados de precipitação: INMET, ANA, CEMADEN
Dados de vazão: ONS e ANA

# Índice Bivariado de Seca (Precipitação-Cota/Vazão) - TSI
## Escala de longo prazo

## Outubro 2026- Parcial



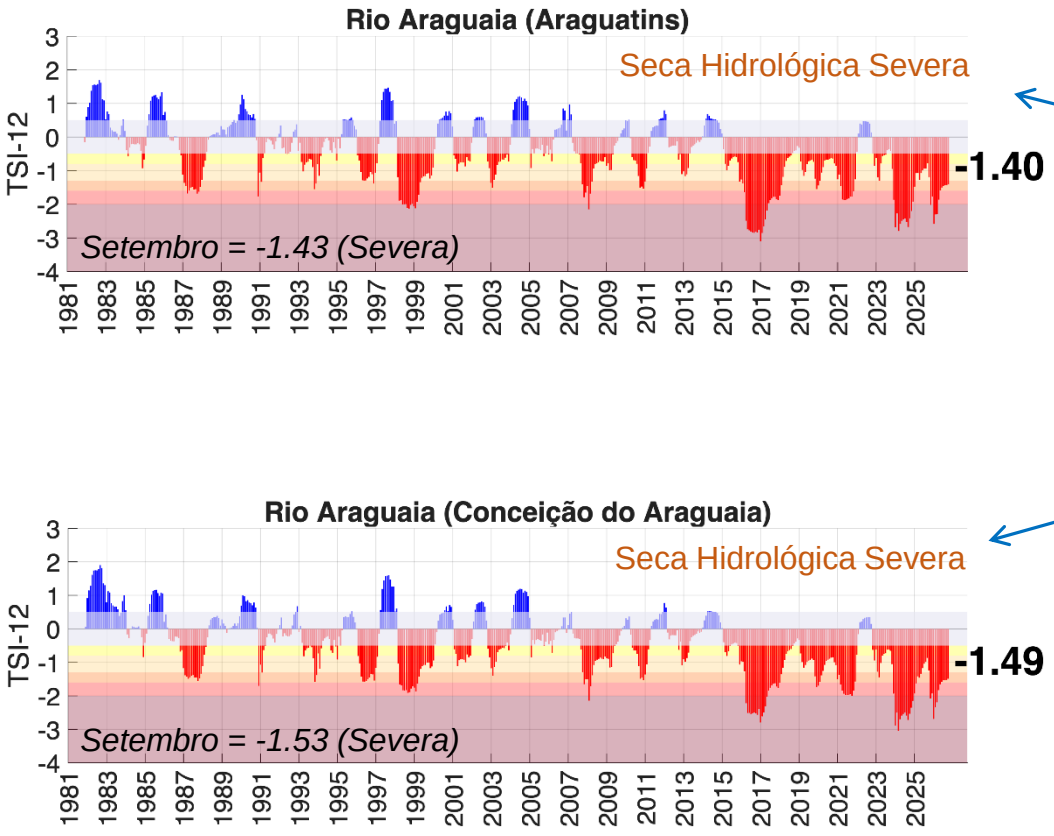



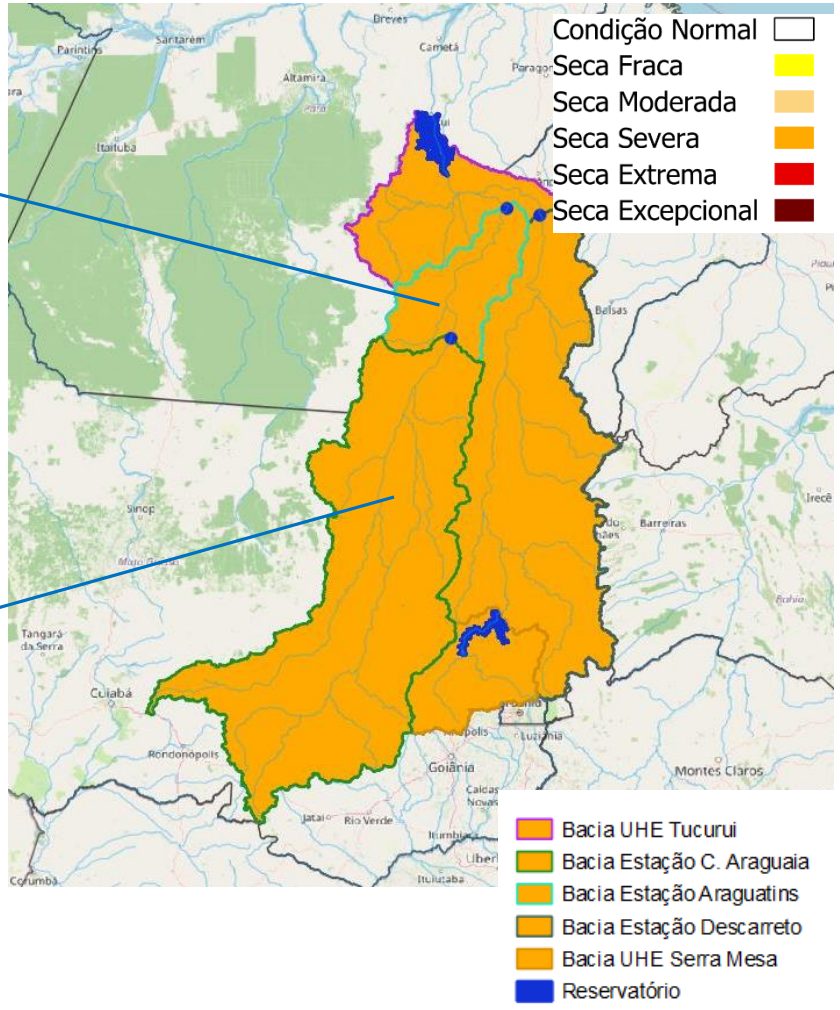

Fonte: CEMADEN
Dados: Precipitação (Cemaden/INMET/CHIRPS) e Vazão (ONS/ANA) - Jan/81-Out/26.

# Índice Bivariado de Seca (Precipitação-Cota/Vazão) - TSI

## Escala de longo prazo

## Outubro 2026- Parcial

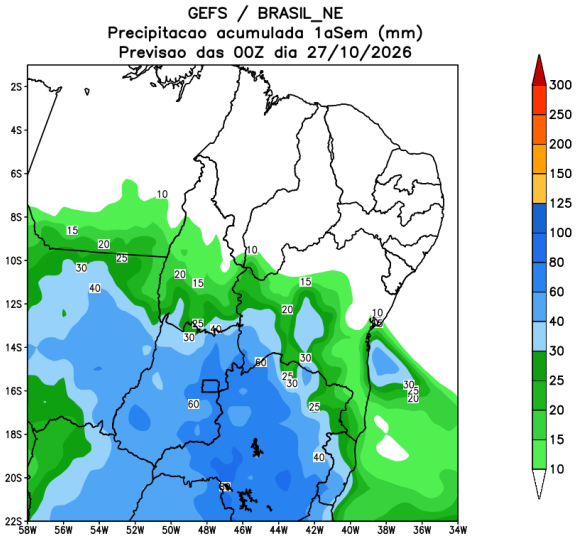


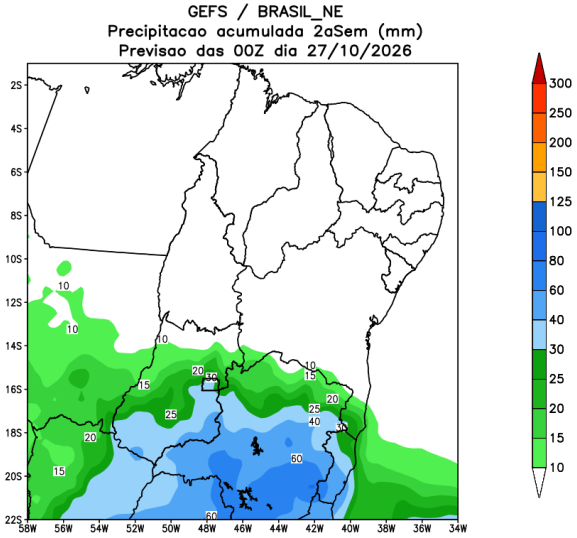




Fonte: CEMADEN
Dados: Precipitação (Cemaden/INMET/CHIRPS) e Vazão (ONS/ANA) - Jan/1981-Out/2026.

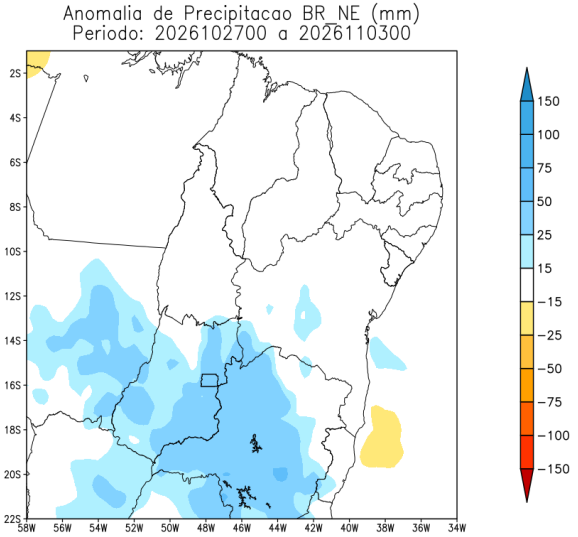
# Previsão de chuva para as duas próximas semanas

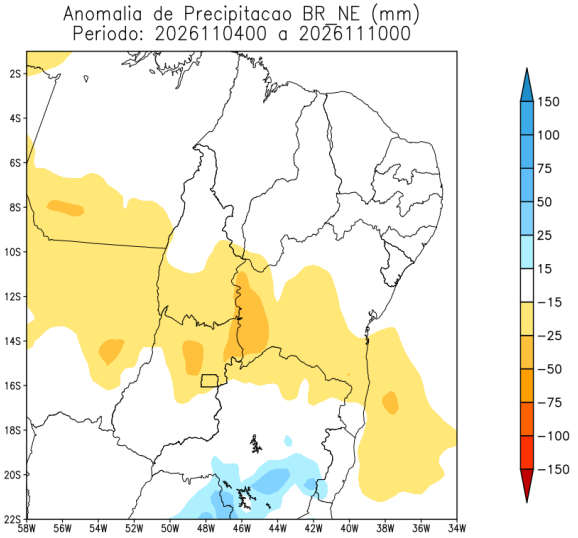
## chuva



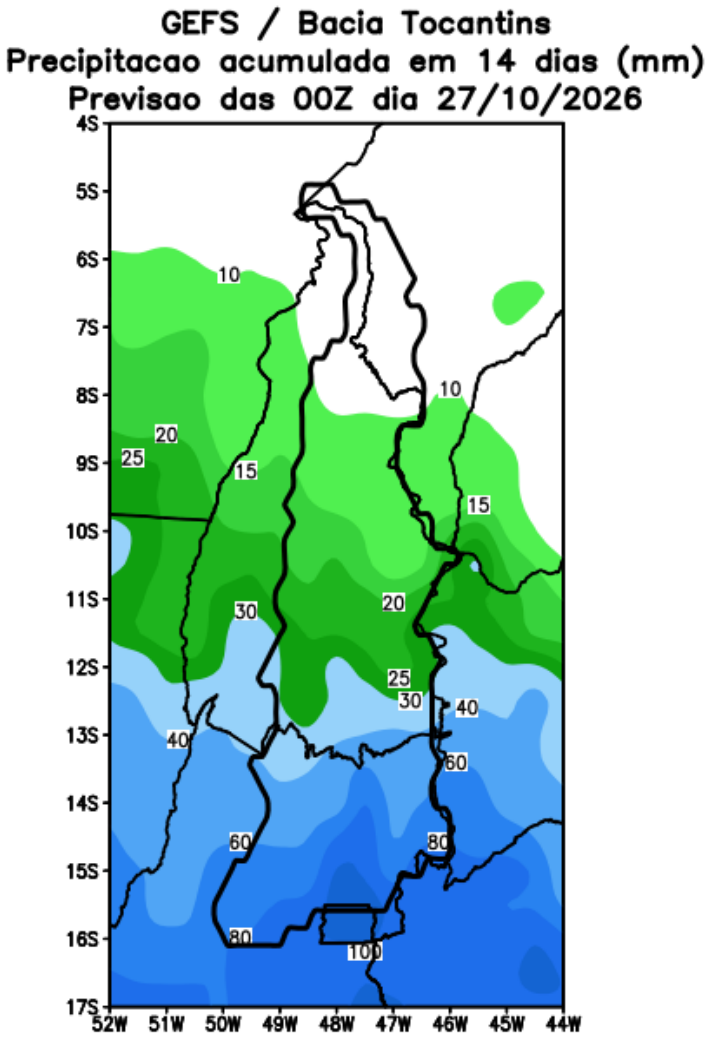

## anomalia

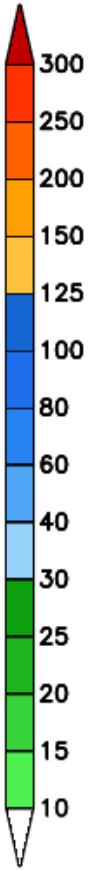


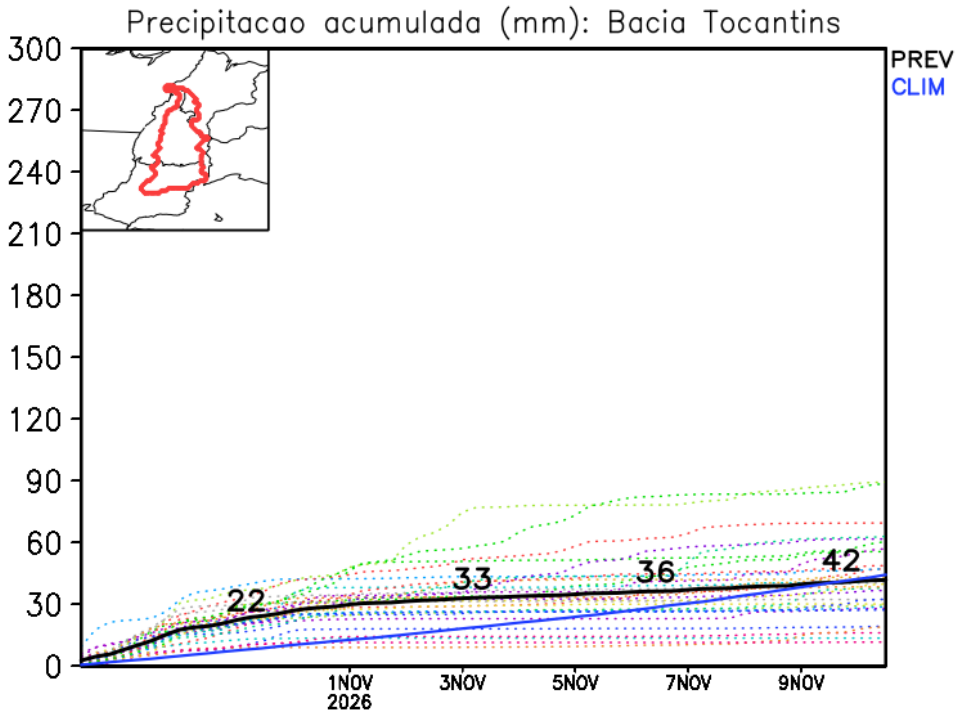




Fonte: GEFS/NOAA

# Previsão Bacia Rio Tocantins



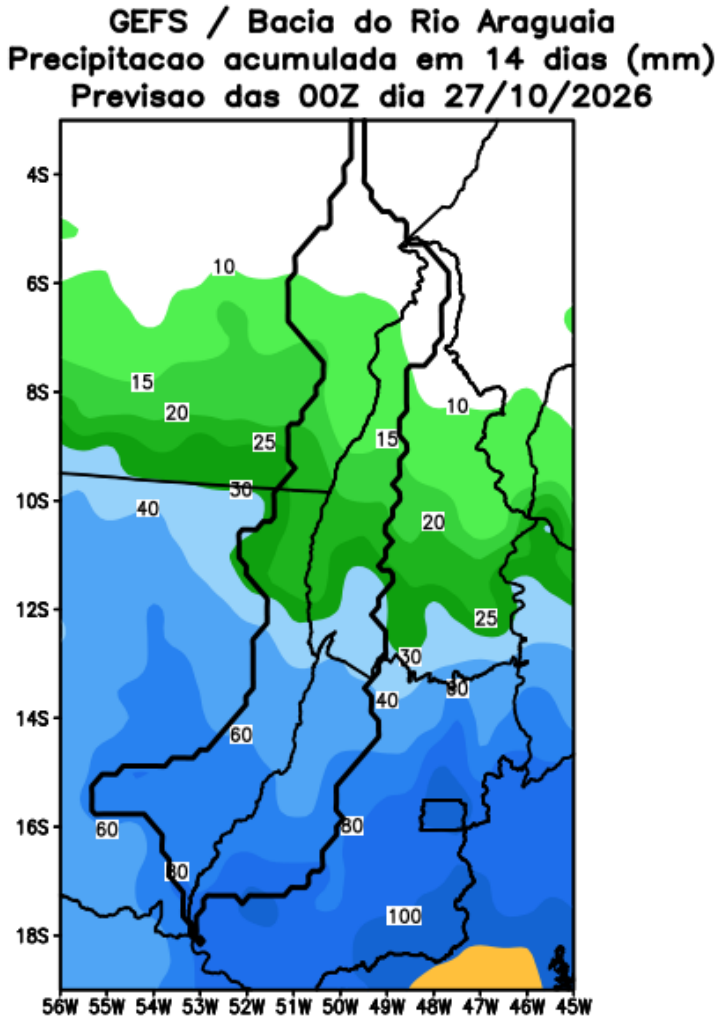



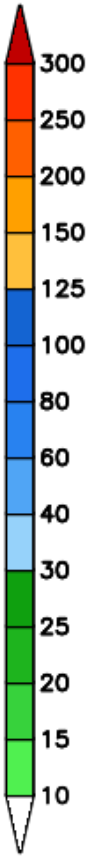



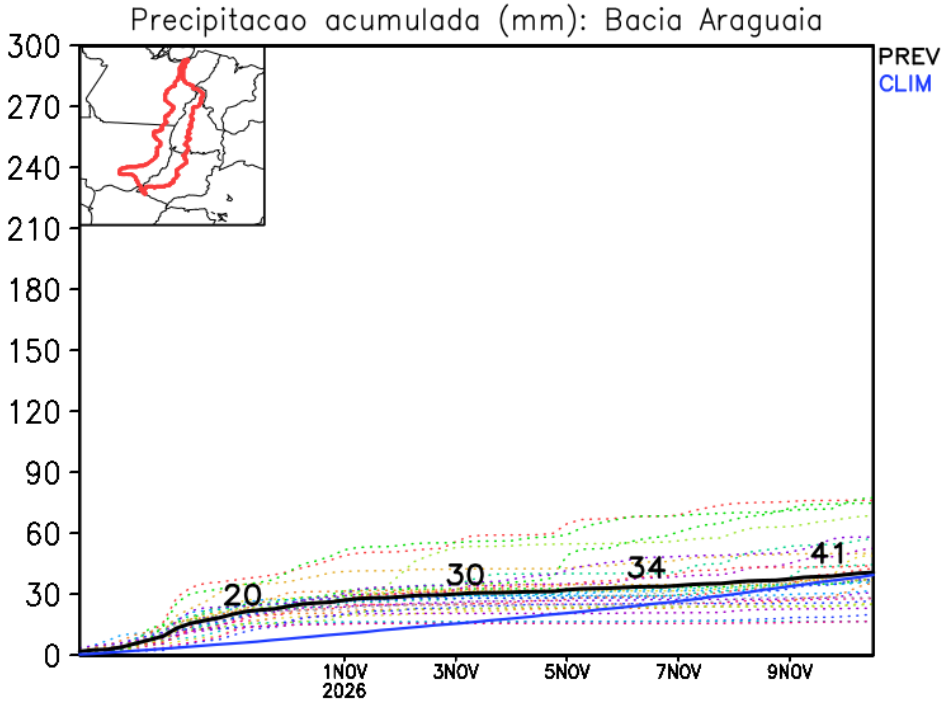



Fonte: GEFS/NOAA

# Previsão Bacia Araguaia

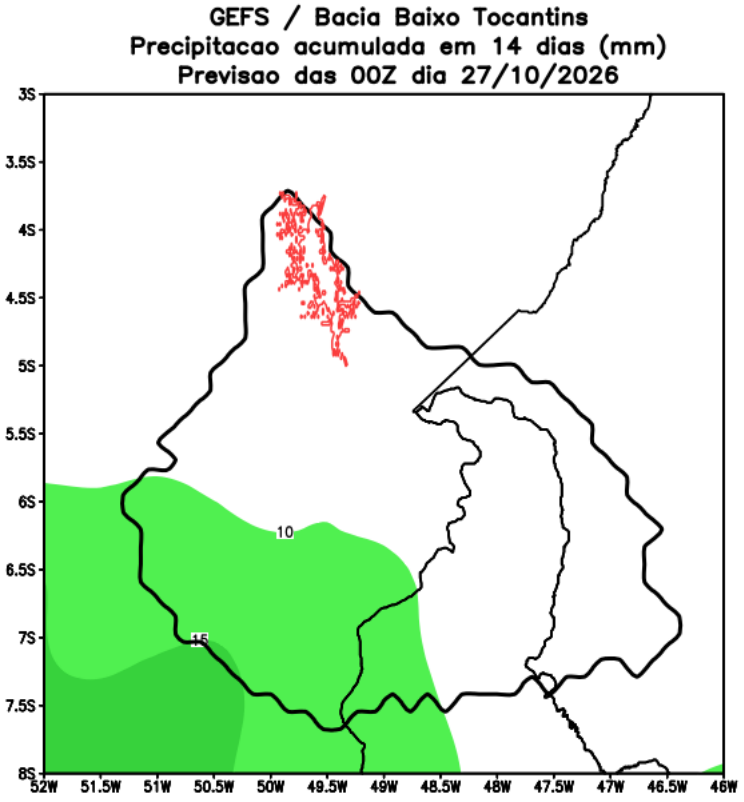


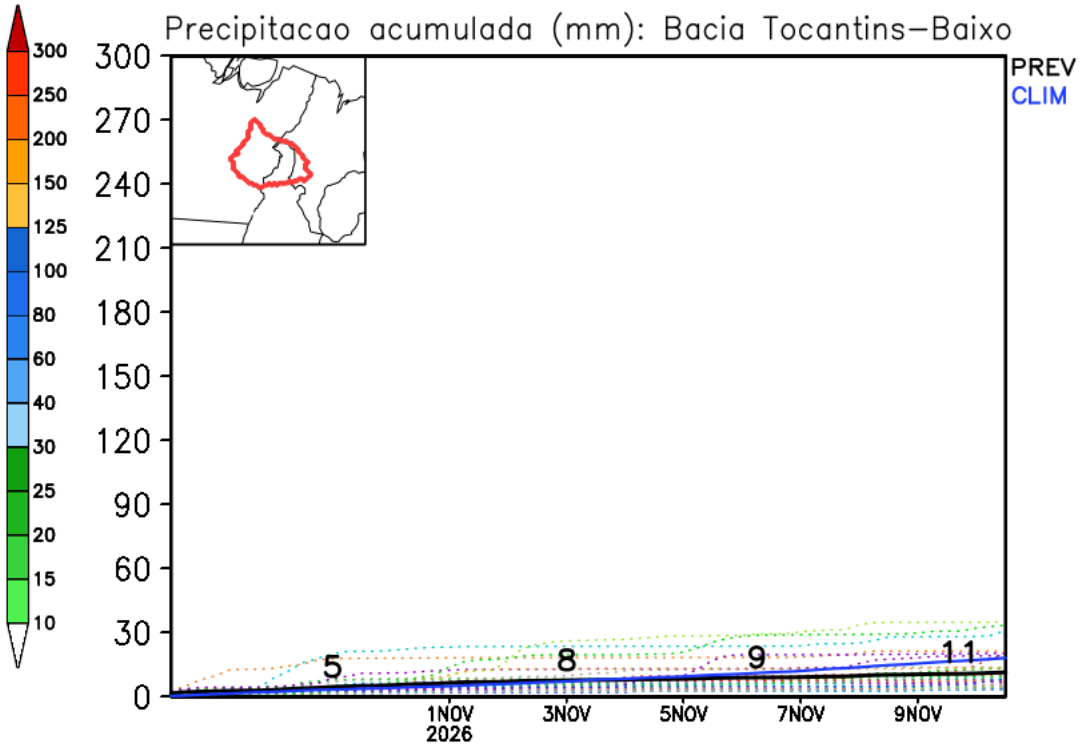




Fonte: GEFS/NOAA

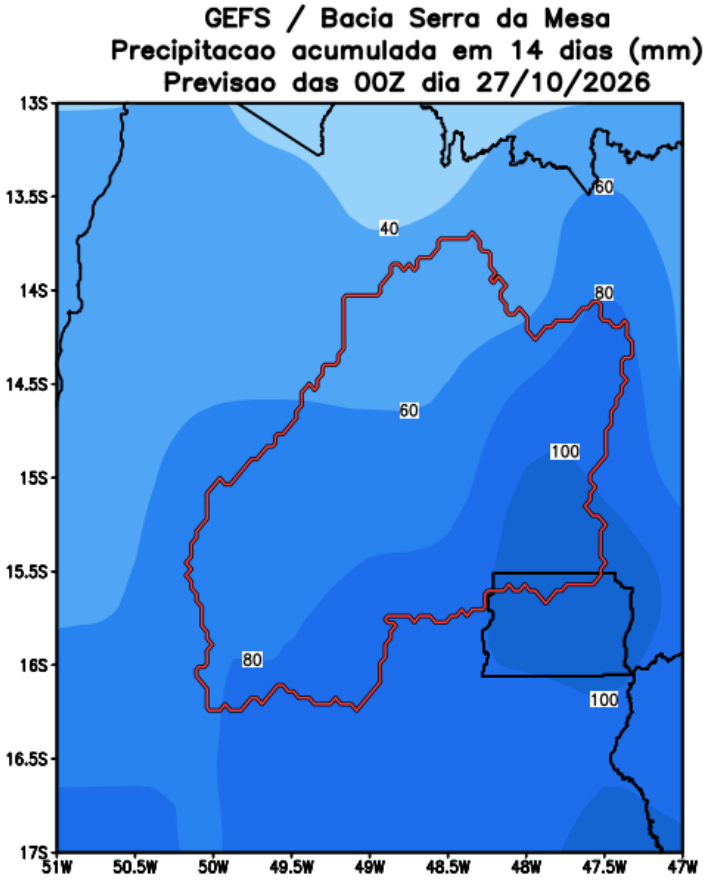
# Previsão Bacia Tocantins (Baixo)

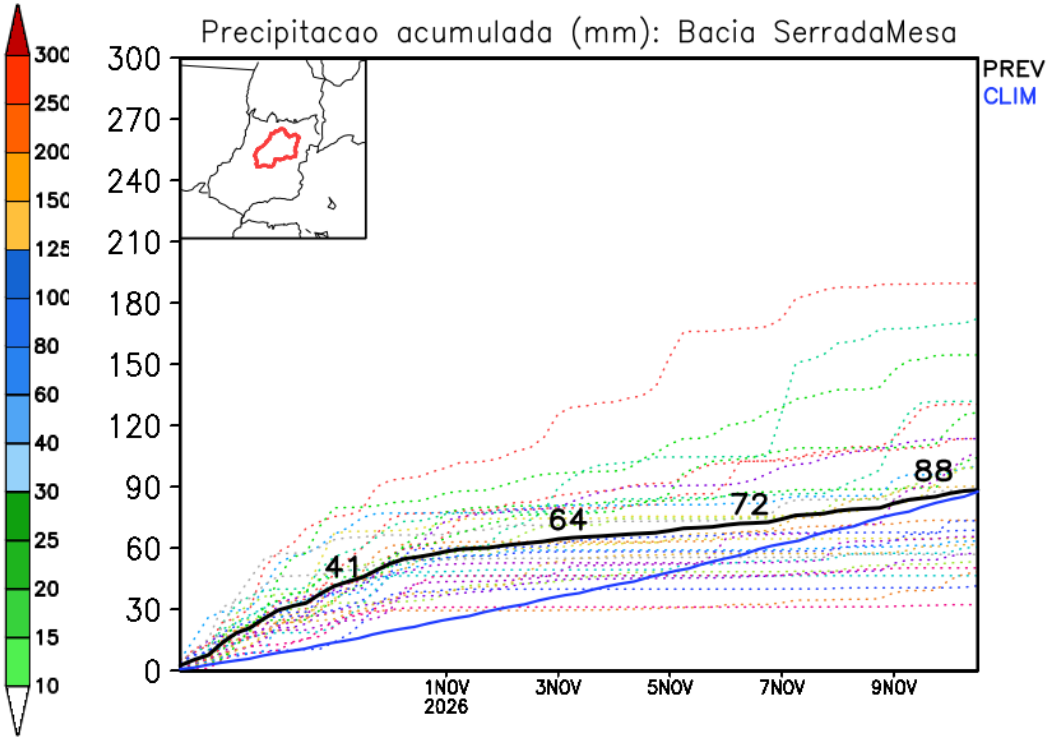



Fonte: GEFS/NOAA

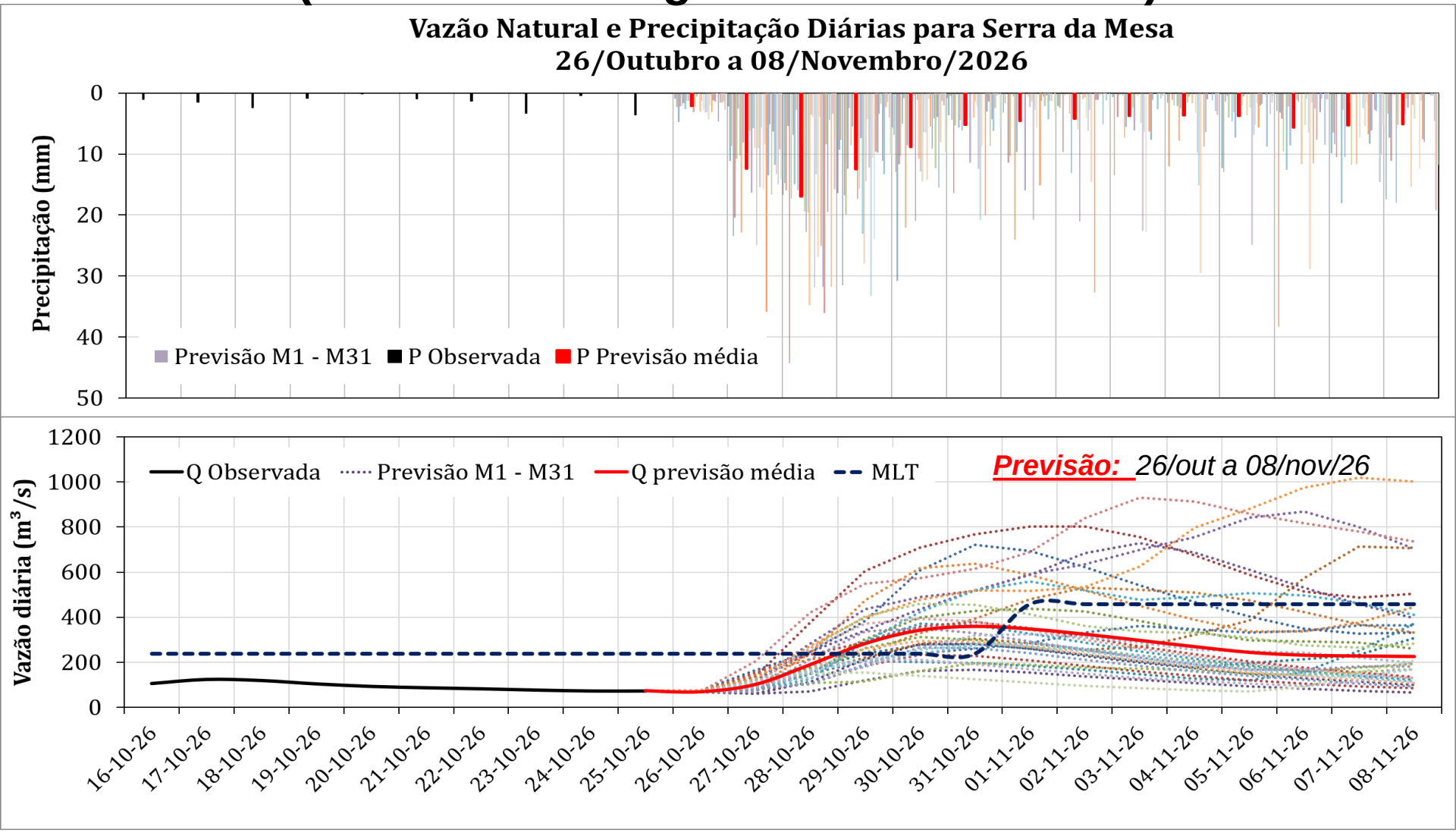
# Previsão Bacia Serra da Mesa




Fonte: GEFS/NOAA

# Serra da Mesa: Previsão de Vazão (modelo hidrológico PDM-CEMADEN)




**Previsão média para os próximos 14 dias: 250 m³/s**

*105% da MLT Out*

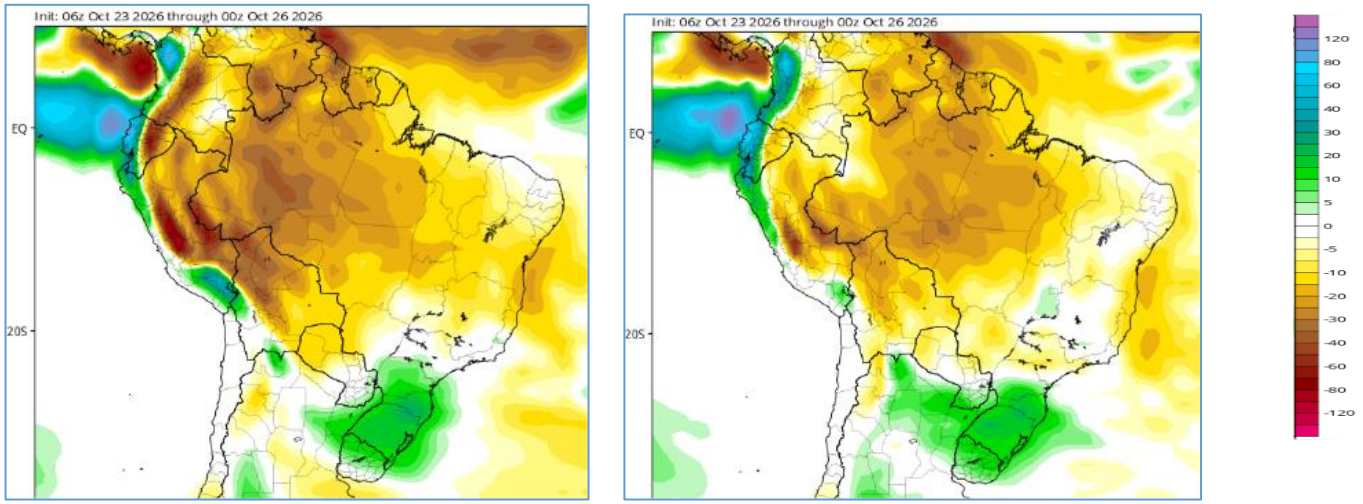
*55% da MLT Nov*

# Tendência 3a e 4a semanas

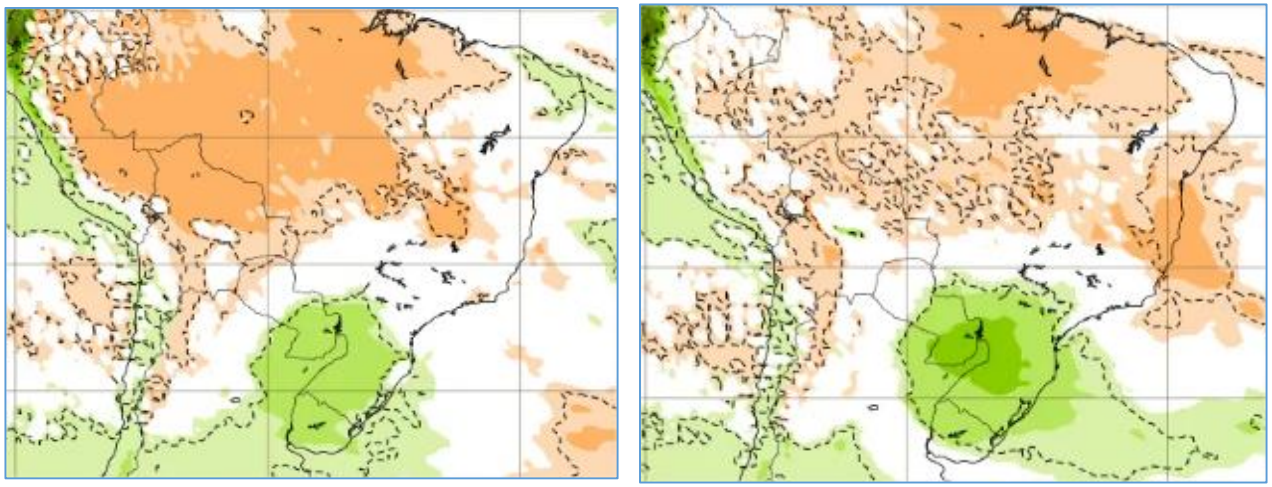
09-16 Nov

16-23 Nov

CFS/NOAA

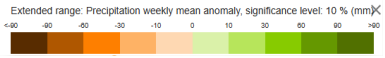



ECMWF



09-16 Nov

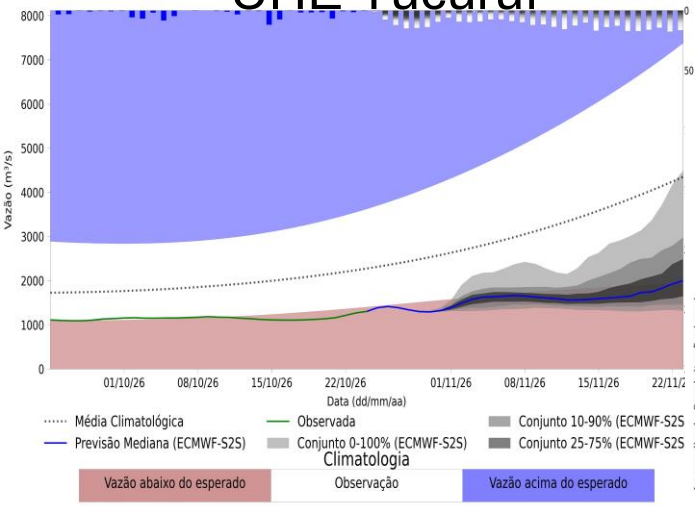
16-23 Nov



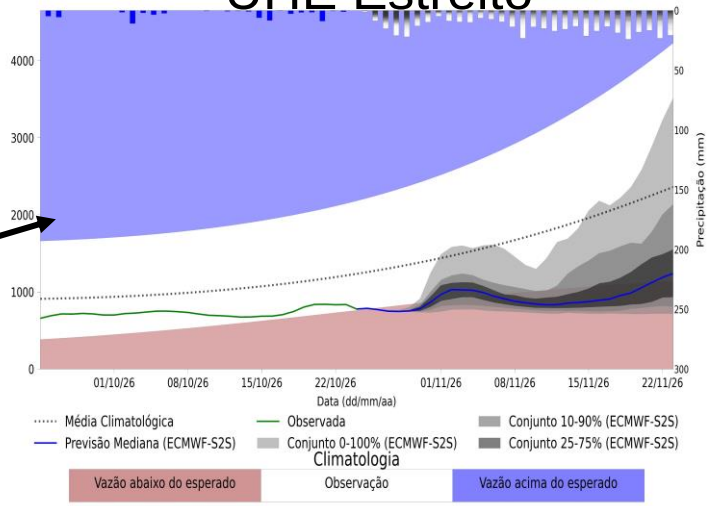
# Previsão de vazão natural na Bacia Rio Tocantins-Araguaia

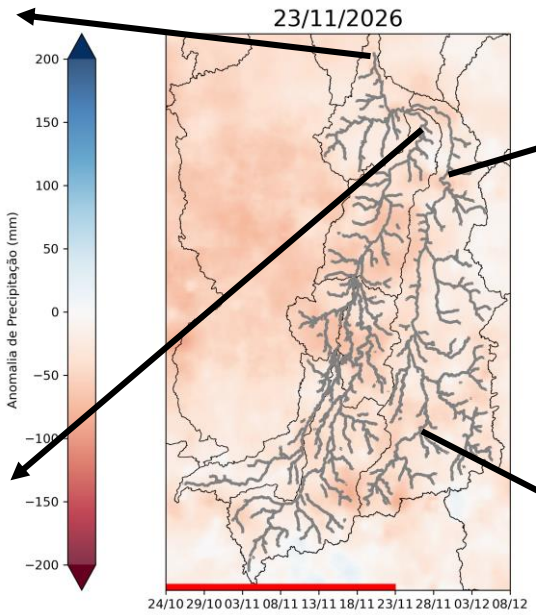
## Previsão: 24/10/2026 a 23/11/2026

### UHE Tucuruí

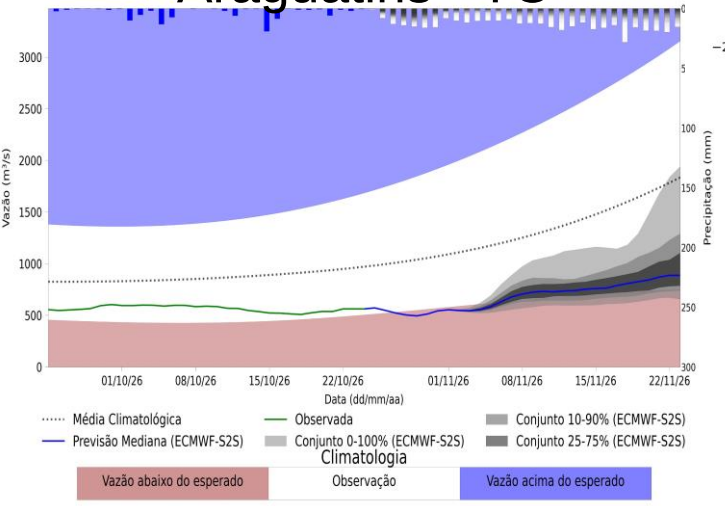



### UHE Estreito

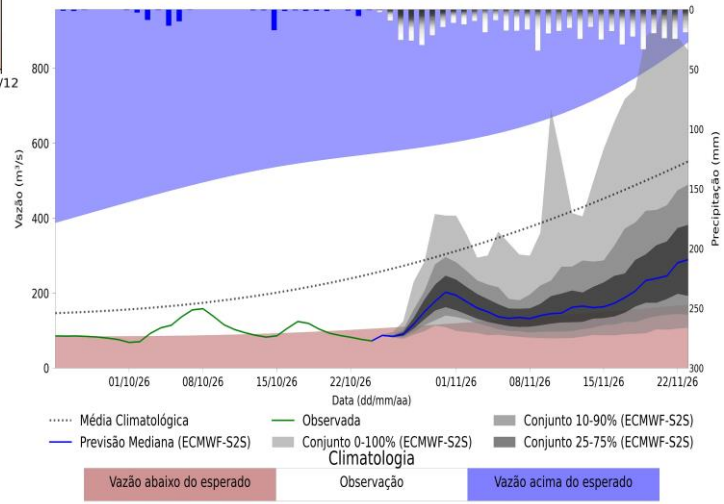



### Araguatins - TO




### UHE Serra da Mesa

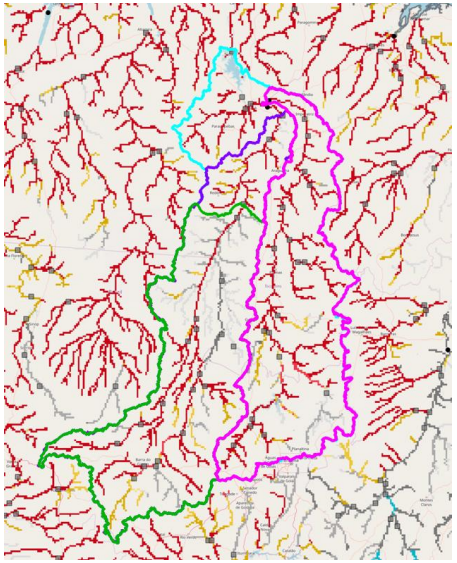



***Fonte:*** *Meteorologia (INMET/MERGE); Vazão (ANA/ONS)*
***MLT:*** *1993-2024*
***Previsão Meteorológica:*** *ECMWF-S2S*

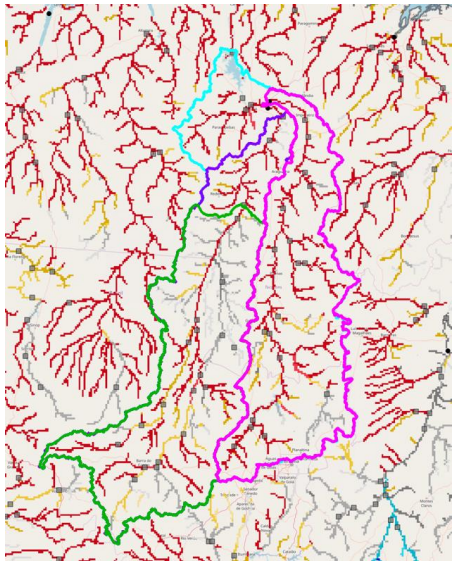
# Previsão Sub-sazonal (45 dias) de vazões naturais para a Bacia do Rio Tocantins (GloFAS)

Previsão: 26/10/2026 – 07/12/2026

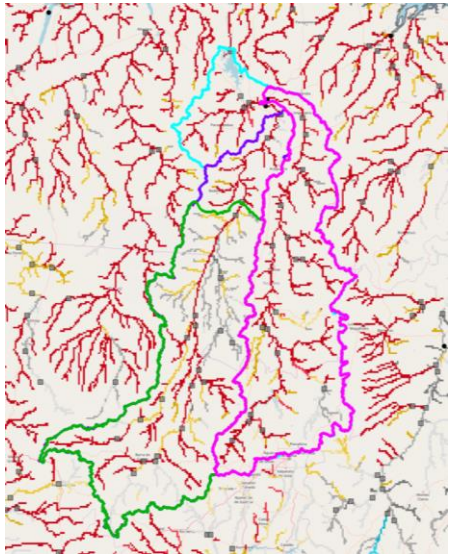
26/10/2026-02/11/2026



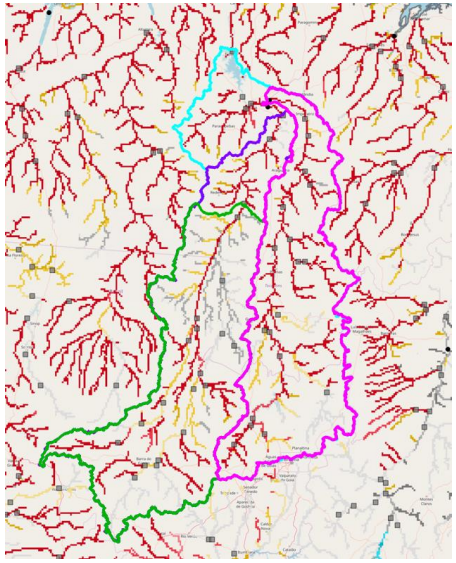
03/11/2026-09/11/2026



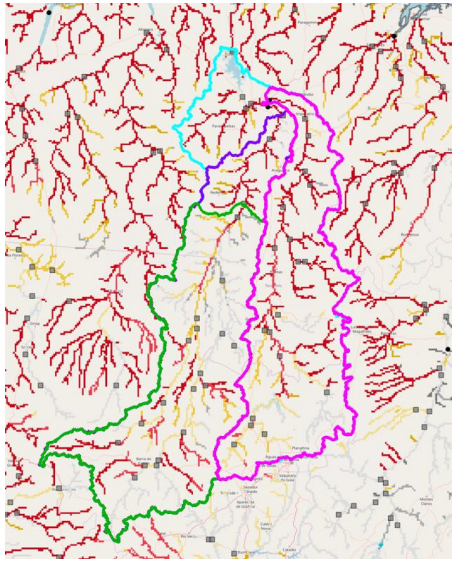
10/11/2026-16/11/2026



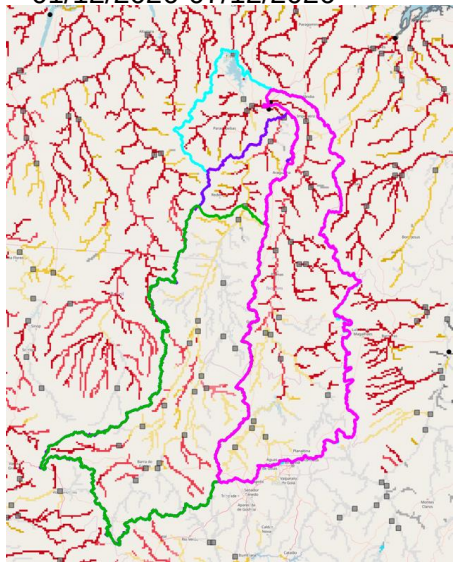
17/11/2026-23/11/2026



24/11/2026-30/11/2026



01/12/2026-07/12/2026



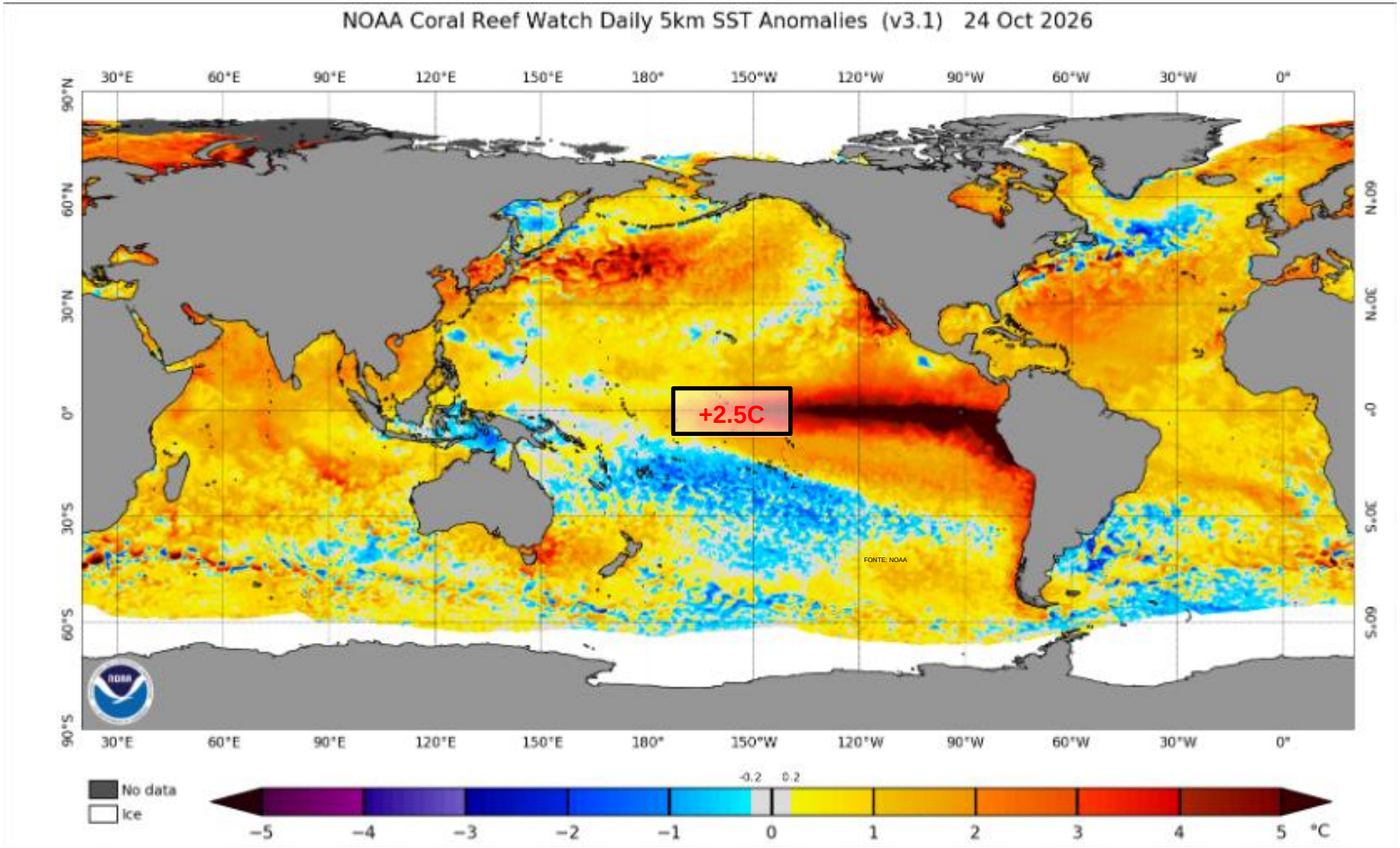
Bacia Tocantins
Bacia Conceição do Araguaia
Bacia Araguatins
Bacia UHE Tucurui

| Categoria de anomalia de vazões (percentil) | Categoria de incerteza | | |
|---|---|---|---|
| | Baixa (0-10) | Média (10-20) | Alta (<20) |
| Muito abaixo (1-10) | | | |
| Abaixo (10-25) | | | |
| Média (25-75) | | | |
| Acima (75-90) | | | |
| Muito acima (90-100) | | | |

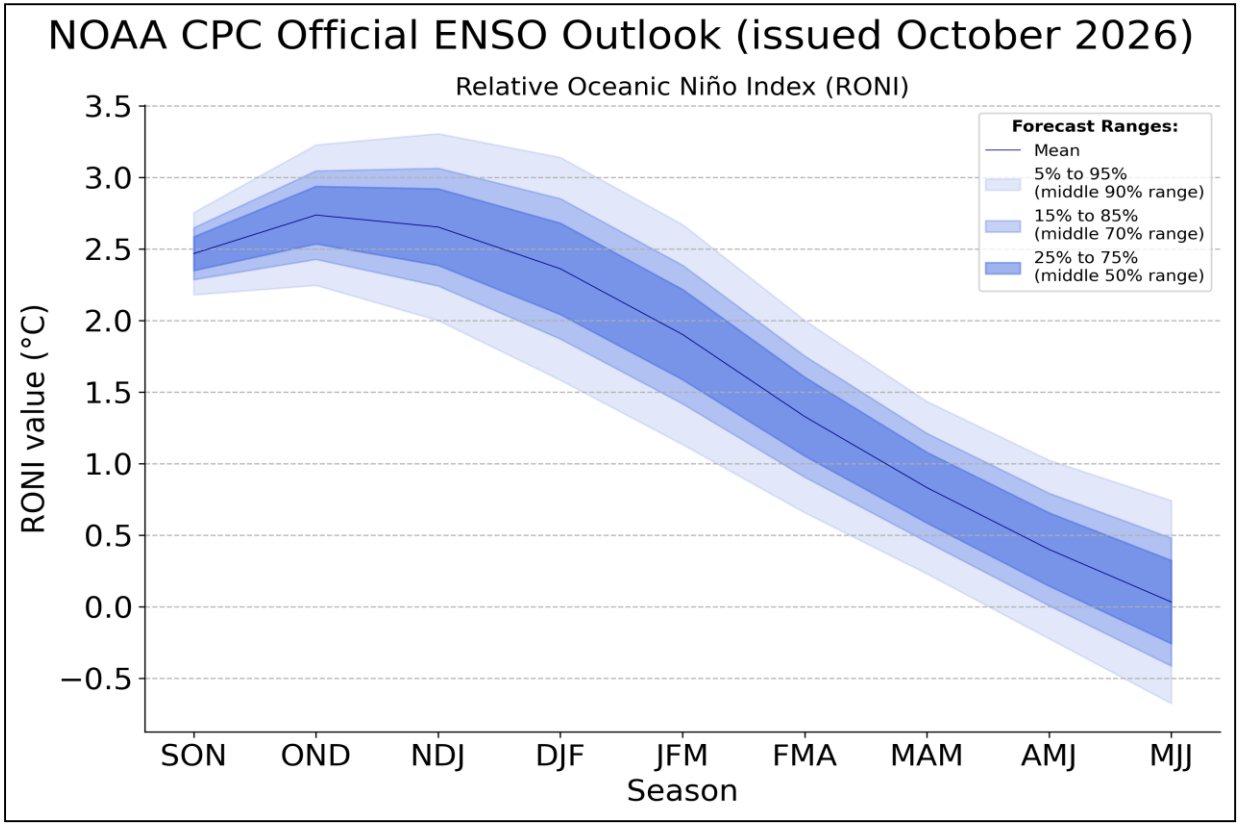
Fonte: Previsão Meteorológica: ECMWF
Previsão de vazão: Lisflood/GloFAS Forecast
https://www.globalfloods.eu/glofas-forecasting/

Cemaden

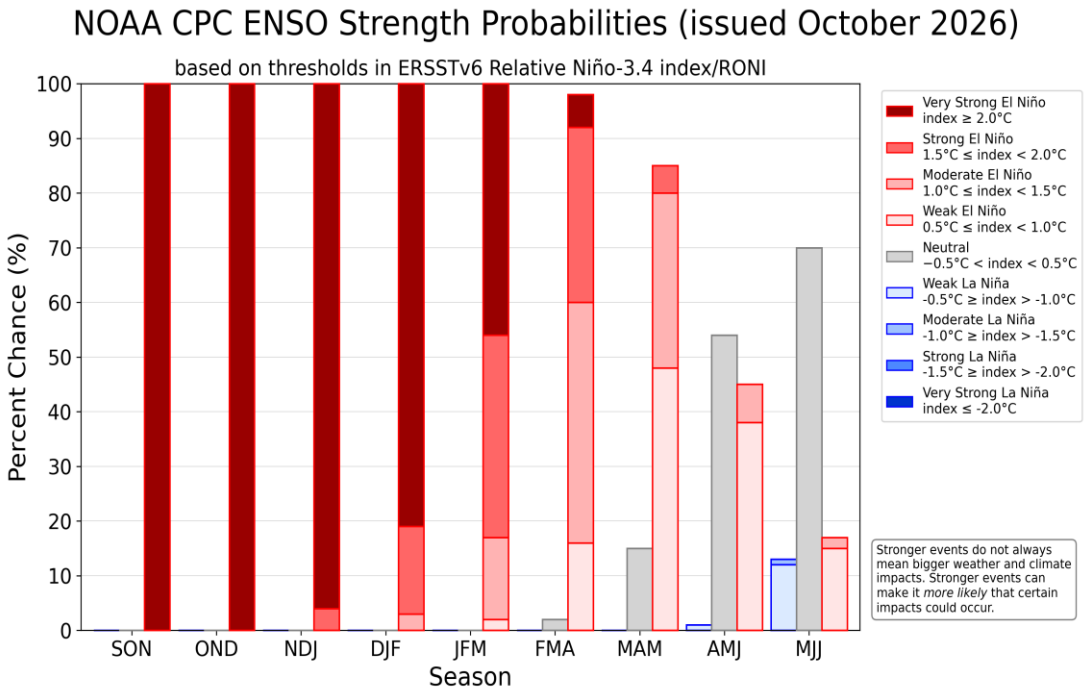
Status Atual: **El Niño**



## Previsão da Intensidade do “ENSO”

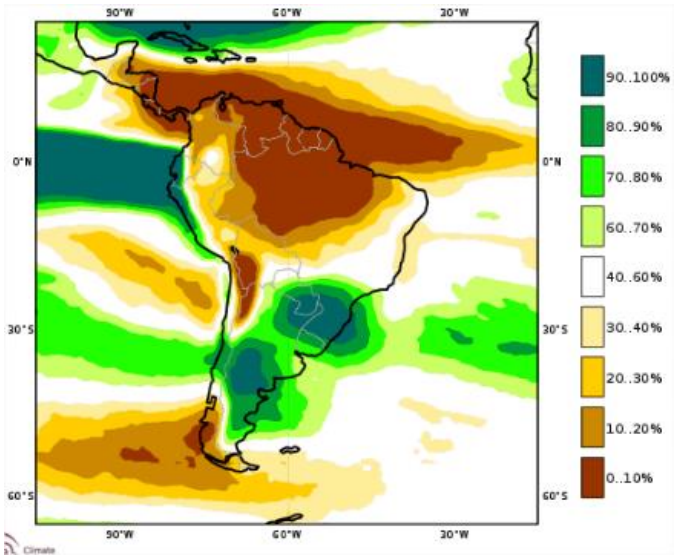


# Previsão da intensidade do “El Niño”

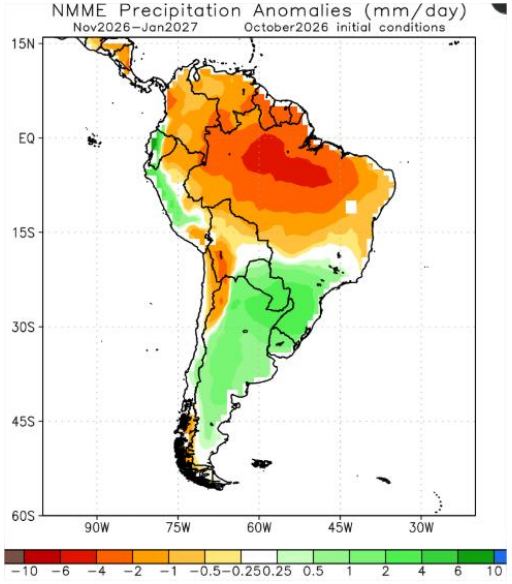


## Previsão Sazonal de Chuva Multi-Modelo

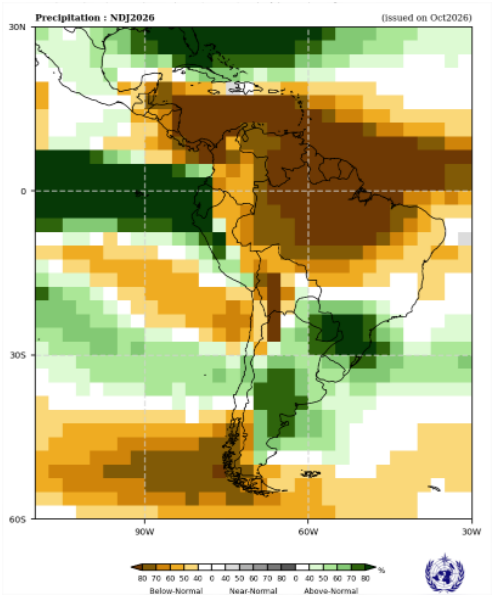
Novembro-Dezembro-Janeiro




Modelos “Europeus”




Modelos Norte Americanos

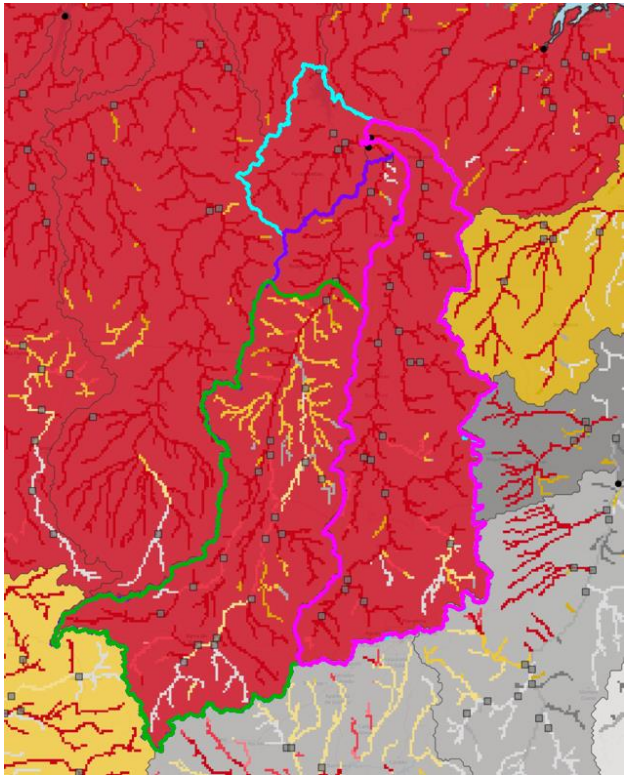



Modelos da WMO

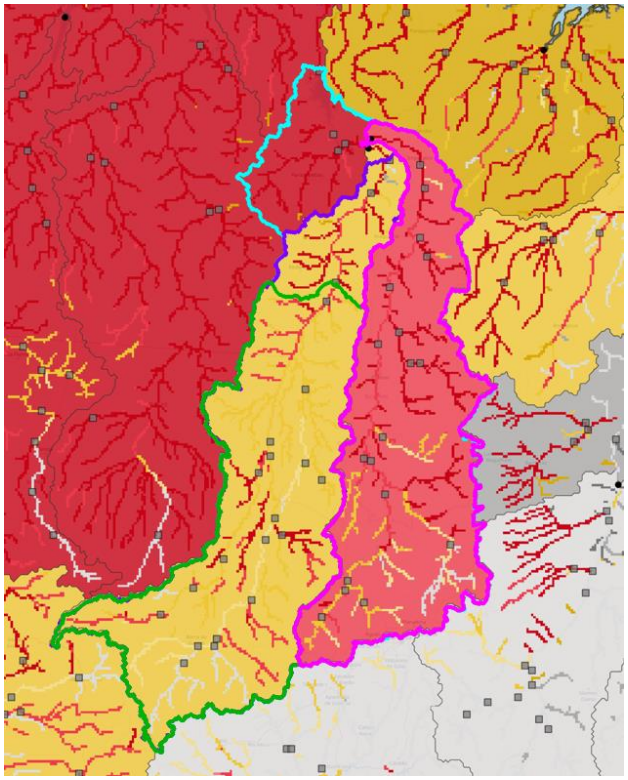
**Previsão Sazonal (DNJ) de vazões naturais para a Bacia do Rio Tocantins (GloFAS)**

Previsão: Novembro, Dezembro e Janeiro de 2027

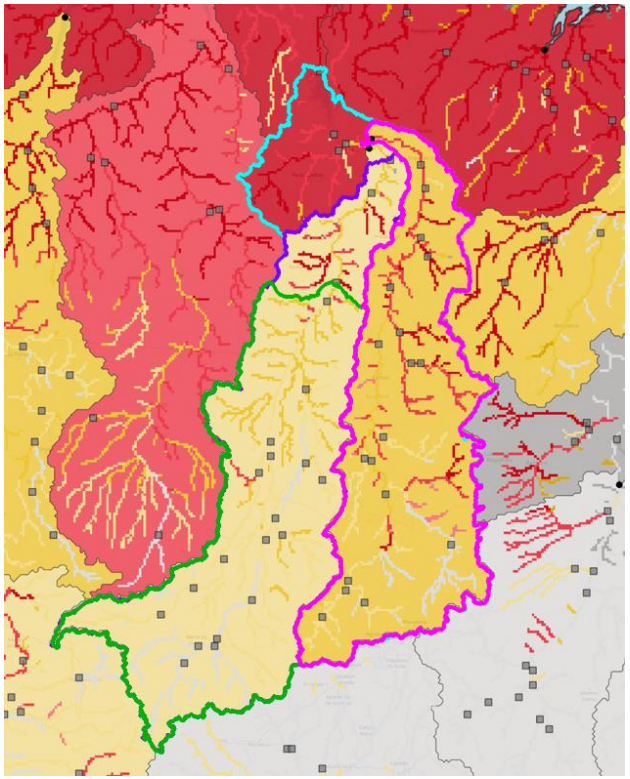
**Novembro**



**Dezembro**



**Janeiro**



Bacia Tocantins
Bacia Conceição do Araguaia
Bacia Araguatins
Bacia UHE Tucurui

| Categoria de anomalia de vazões (percentil) | Categoria de incerteza | | |
|---|---|---|---|
| | Baixa (0-10) | Média (10-20) | Alta (<20) |
| Muito abaixo (1-10) | | | |
| Abaixo (10-25) | | | |
| Média (25-75) | | | |
| Acima (75-90) | | | |
| Muito acima (90-100) | | | |

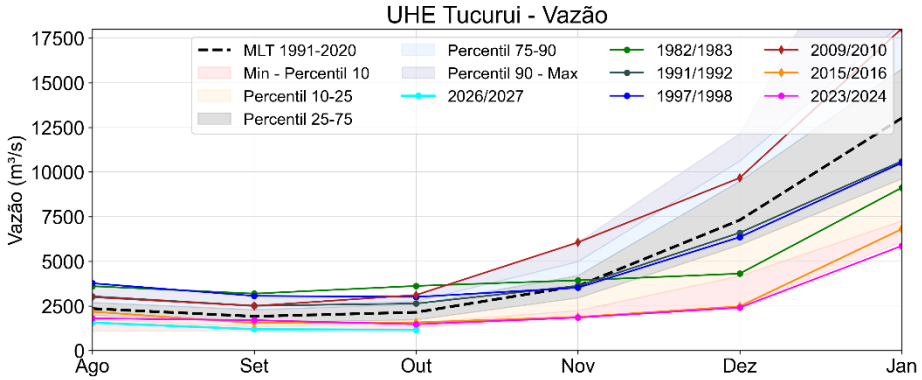
Fonte: Previsão Meteorológica: ECMWF
Previsão de vazão: Lisflood/GloFAS Forecast
https://www.globalfloods.eu/glofas-forecasting/

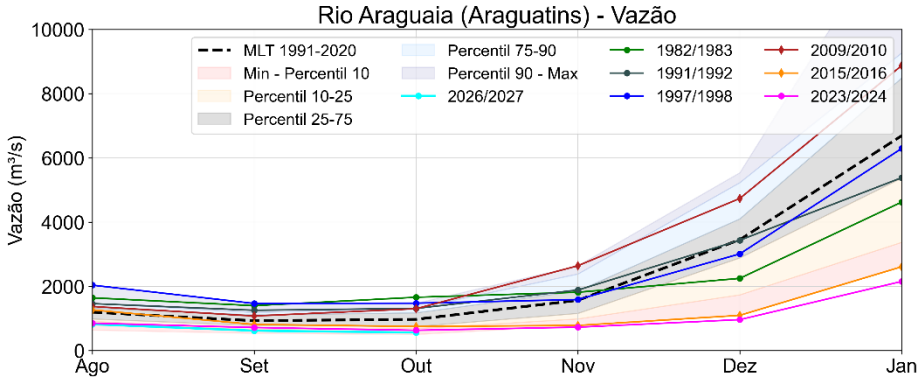
Cemaden

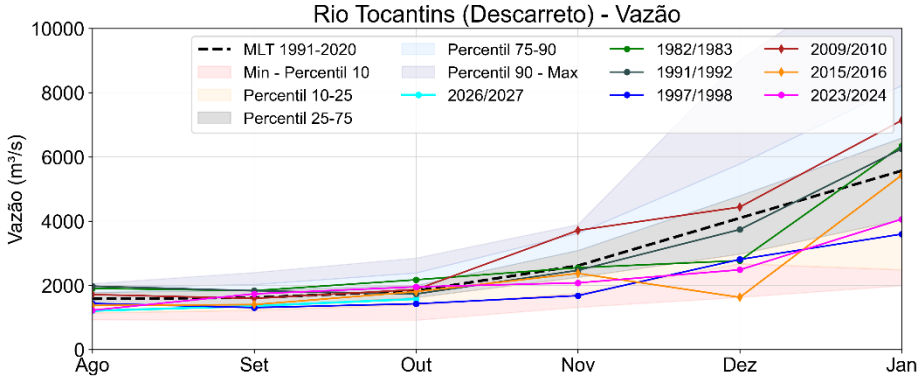
# Anos Análogos de El Nino

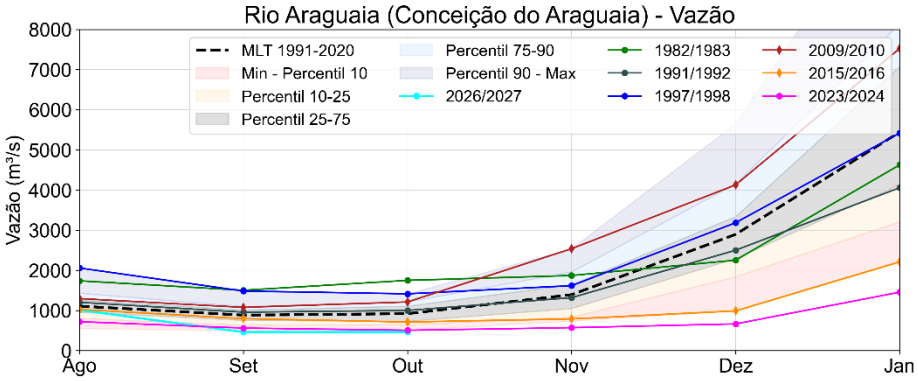
## Trimestre NDJ

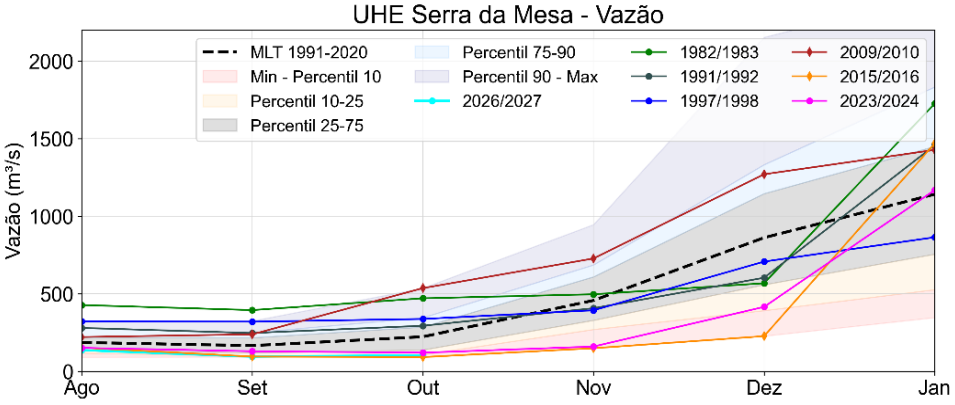
- 1982/1983
- 1991/1992
- 1997/1998
- 2009/2010
- 2015/2016
- 2023/2024



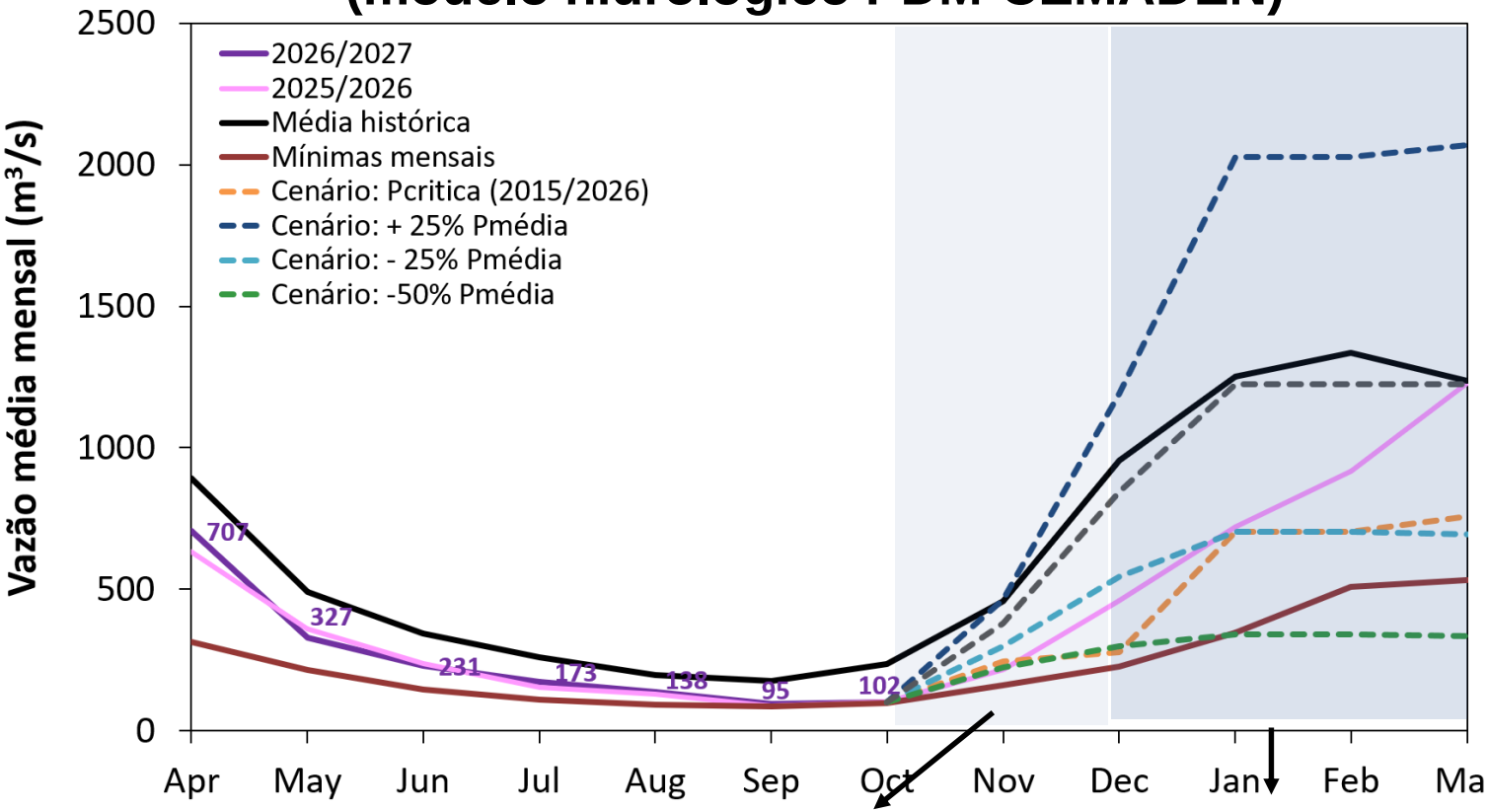




Fonte: CEMADEN/MCTI Dados: Vazão (ONS/ANA): 1982-2024

# Serra da Mesa: Projeção de Vazão (modelo hidrológico PDM-CEMADEN)




**Nov e Dez (Meses Chuvosos)**

**MLT: 705 m³/s**

| Vazão | % MLT |
|---|---|
| 828 m³/s | 117% |
| 611 m³/s | 87% |
| 422 m³/s | 60% |
| 261 m³/s | 37% |

**Jan-Mar (Meses Chuvosos)**

**MLT: 1275 m³/s**

| Vazão | % MLT |
|---|---|
| 2042 m³/s | 160% |
| 1224 m³/s | 96% |
| 700 m³/s | 55% |
| 722 m³/s | 57% |